Course Overview

Internet Infrastructure Security

CIS0012

Nice to meet you

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Education

Ph.D. in Seoul National University
B.S. in Seoul National University

Experience

Postdoc at University of Virginia
at Seoul National University

Expertise

Network security, Data-driven security analysis, …

Course information

- Meet once a week
  - Wednesday 15:00 – 18:00

- Textbook
  - Not required
  - Lecture slides are provided (usually before the class starts)
    - https://nestlab.hanyang.ac.kr/teachings/internet-infra-security/
  - Research papers

Course information

- Announcements
  - via LMS (Hanyang Univ. Portal)
  - Ex) class cancellation, makeup class, grade announcements, etc.

- Grading
  - Attendance: 10%
  - Midterm exam: 45%
  - Debate (Paper presentation): 45%

Course policies

- Attendance
  - You must attend at least 2/3 of all classes to be eligible to take the exams
  - Absence/tardiness still counts as attendance if you submit an OFFICIAL "**Excused absence form (공결신청서)**" from the administration office

- Exams
  - Missing an exam or being caught cheating results in CANCELLANTION of your grade
  - No leaving your seat during an exam, including for restroom use – Only allowed after submitting your answer sheet

Course etiquette

- Basic etiquette
  - Please refrain from eating during class
  - Water/drinks are fine, just don't let them disrupt class
  - You may use the restroom during class, but only when truly necessary and without disrupting others
  - Please refrain from non-class-related behavior (sleeping at your desk, chatting, etc.)

- On electronic device use
  - Use your phone only when truly necessary (take urgent calls outside)
  - Laptops/tablets may be used during class
    - Only for class-related purposes such as viewing lecture materials or taking notes
    - Please refrain from non-class use (games, videos, music, etc.)

Contact

- Email
  - Instructor (Hyeonmin Lee): hyeonminlee@hanyang.ac.kr

- Email policy
  - Include [Internet Infra Security] in your email subject line
    - Example: “[Internet Infra Security] Question”
  - In the body, introduce yourself (name/department/student ID) before getting to your question
  - Otherwise, there might be cases that I couldn’t answer it (ex. your email can go to spam folder)

Semester at a glance (Tentative)

Week

01. Course Overview
02. PKI and TLS
03. DNS Security
04. BGP and Email Security
05. Paper Presentation (1st round)
06. Paper Presentation (1st round)
07. Own Research Presentation
08. Paper Presentation (1st round)

Week

09. Midterm Exam
10. Paper Presentation (2nd round)
11. Paper Presentation (2nd round)
12. Paper Presentation (2nd round)
13. Paper Presentation (2nd round)
14. Paper Presentation (2nd round)
15. Paper Presentation (2nd round)
16. Make-up Class (if necessary)

Why presentation?

- Presenting is a core competence for a graduate student
  - Your work is worth what you can get other people to understand of it
- And it stays core after you graduate
  - A design review, a conference, a job talk … The same skill!
- This is why the Debate (paper presentation) component is 45%
  - The same as the exam
- You will present twice
  - The second one is worth more than the first

How the presentations are assessed

	First round	Second round
Weeks	5, 6, 8	10 to 15
Per student	Total 30 min : 20 present + 10 discussion	Total 60 min : 40 present + 20 discussion
Assessed on	• Did you read and understand the paper • What did you build the presentation out of • How effectively did you deliver it	The same three
Your questions	At least 2 qualifying questions	At least 4 qualifying questions

Two things about the discussion

- **[When you are presenting]** you are judged on the honesty of your answer, not the difficulty of the question
  - If you don't know the reason, you can offer a hypothesis or your own reasoning about it

- **[When you are not presenting]** your questions count
  - At least 2 in the first round, at least 4 in the second round
  - A qualifying question is about the paper’s method, its limits or one of its claims
  - Asking what a term means is welcome, but does not count

What makes a good paper to present

- From a top-tier venue (ex. IEEE S&P, USENIX Security, CCS, NDSS, and similar)
  - A venue is where research is published; top-tier means its reviewing is the strictest
  - You are borrowing their peer review, because you cannot assess soundness alone
- Recent enough to describe the Internet as it is now
- Clearly tied to one of our course topics: PKI, TLS, DNS, BGP, email

Week 7 – Your own research

- Record a 20-minute video presenting your own research topic, and submit it

- There is no class meeting
- Submitting the video is what counts as your attendance for that week

- It is about your research, not about this course topics

Example: You visit a website

One action, four systems, in the order they actually happen

The Internet is made of many underlying systems

- The Internet you use is not one thing. It is many systems working together
  - Each one was built for a different job, by different people, at a different time

- A protocol is an agreed set of rules for how two machines talk to each other
  - Both sides have to follow the same rules, or nothing works

- Most of these systems are invisible when they work
  - You only meet them when something breaks

- This course opens four of them

The four topics of this course

- Border Gateway Protocol (BGP)
  - How data finds its way across the Internet
- Domain Name System (DNS)
  - How a name you type becomes a machine you reach
- Public Key Infrastructure (PKI) and Transport Layer Security (TLS)
  - How you know the machine you reached is the right one
  - These three are infrastructure: everything else is built on top of them
- Email
  - Not infrastructure, but the most common way people communicate

The rest of today: One story

- We follow one ordinary action, step by step

- You open a browser and visit a website (ex. *example.com*)

You type a domain name. Now what?

- You type *example.com* and press enter
  - That string of letters is called a domain name

- Your browser now has to send a request to a machine somewhere in the world

- But it does not know which machine
  - It has a name
  - But It does not have a location

- So before anything else can happen, something has to answer one question:
  - Where is *example.com*?

The machine needs a number, not a name

- *Names* are for people. *Machines* find each other by number

- An IP address is the number a machine is reached at
  - Something like 1.2.3.4 – every machine on the Internet has one

- The name and the number are not related to each other
  - You cannot work one out from the other, any more than you can work out
  - A phone number from a person's name

- So somebody has to keep a list, and somebody has to answer questions about it

So something has to translate: Domain Name System (DNS)

- DNS is the system that turns a domain name into an IP address
  - That translation is the first reason DNS exists

- Actually, your browser does not do the lookup itself
  - It asks somebody

- The entity that looks a name up on your behalf is called a *resolver*
  - Usually run by your university, your internet provider, or a public service
  - Not by you

How DNS answers: nobody knows everything

The name space is all the names; delegation hands part of it to somebody else

Root (.)
knows who runs *com.*

↓ delegates

com.
knows who runs *example.com.*

↓ delegates

example.com.
knows the address

↓ answers

1.2.3.4
(IP address)

DNS hierarchy

But that question has to travel somewhere

- We just said your resolver asks the *root (.)*, then *com*, then *example.com*
- Those are machines in other buildings, in other countries
- So the question itself has to cross the Internet to reach them
  - And so does the answer, coming back
- Which raises a second question:
  - *How does anything get from here to there at all?*

Your data travels in packets, through networks you do not control

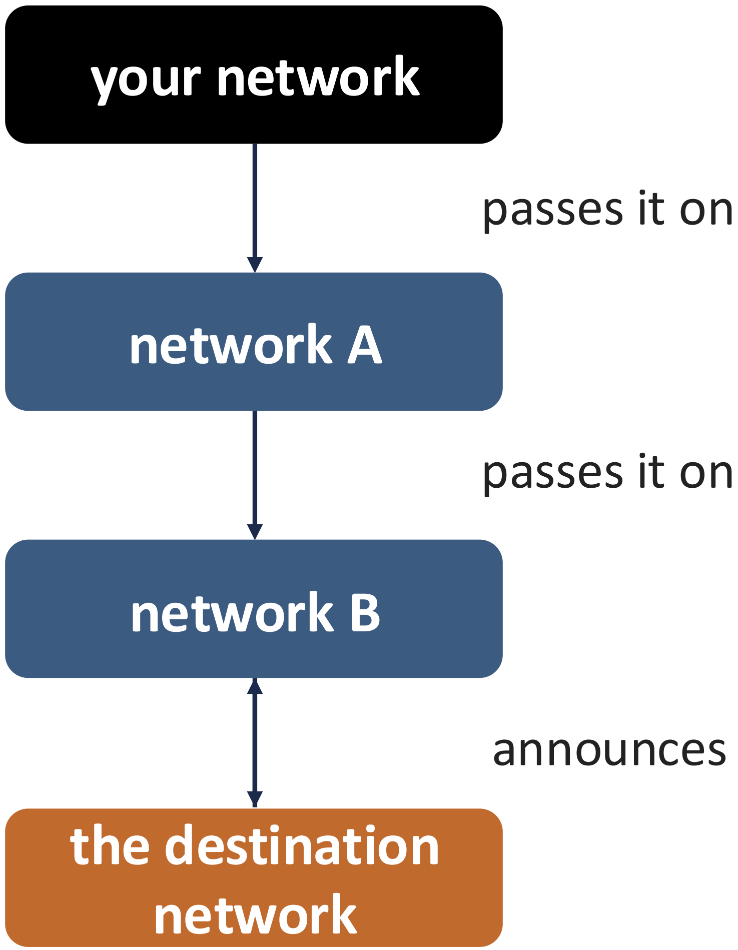
- Data does not travel as one piece – It is cut into “packets”
  - A packet is a small unit of data with an address written on it

- Each packet is passed from network to network until it arrives
  - Ex) University's network → other network → another → another …

- You do not own any of them, and you did not choose any of them
  - Which means somebody has to decide, at every step, which way to send it

Who decides where a packet goes: Border Gateway Protocol (BGP)

Routing path



Announcement?
send me packets for
these addresses

The answer comes back. You connect to the server

- The DNS question travelled by BGP. The answer came back the same way
- Your browser now has an IP address, and sends its request there
  - That request travels by BGP too, over the same kind of path
- A server answers, and a page appears
  - A machine whose job is to answer requests
- Two of the four systems have now done their work
  - And here is the question neither of them can answer

How do you know it is the right server?

- Something answered at that address.. How do you know it is really *example.com*?

- **DNS** told you an address; It did not prove the address was correct

- **BGP** delivered your packet; It did not check where it ended up

- Whatever answered could be:
  - Real server
  - or Somebody who got your packets instead, and answered in its place

- Nothing you have seen so far can tell those two apart

Somebody has to vouch: Certificates and CA

- The server sends you a certificate
  - A certificate is a document saying "*this name belongs to this machine*"
  - "Signed" by somebody else, so you can check it was not written by an impostor
- The party that signs/issues it is a Certificate Authority (CA)
  - An organization your browser believe
- The system of CAs and certificates is called Public Key Infrastructure (PKI)
  - A *public key* is a value a machine publishes so anyone can check what it signed
- Transport Layer Security (TLS) is the protocol that uses it on a live connection

Sending email goes through much the same process

- For example, you send an email to somebody@example.com
- The name after the @ has to become a machine – **DNS**
- The message travels in packets across networks nobody coordinates – **BGP**
- The receiving server has to decide whether to believe the message really came from you

Topics – Summary

In the story	The need	The system
You type a name	A name is not a location. Something must translate it	**DNS**
The question must travel	No one owns the path. Something must decide where each packet goes next	**BGP**
Something answers	Anyone could answer. Something must prove it is the right party	**PKI/TLS**
You send a message	The same three needs, in a different order, with a different ending	**Email security**

Are these safe? No.

- Everything you do online rests on those four
  - banking, messaging, medical records, government services

- So it would be reasonable to expect them to be carefully secured

But they are not ...

- DNS: does not check whether an answer is true
- BGP: does not check whether an announcement is true

Why?

- Most of these protocols are decades old
- They were designed when security was not a consideration
- They were built by a small research community whose members knew each other
  - The problem to solve was ‘does it work at all?’
  - Not ‘what if one of us is lying?’
- So there is no security in the basic design
- The Internet then became something nobody designing it had planned for ...

So security was added later

- Security was added afterwards, on top of protocols that did not have it
  - A signature added to DNS answers, a way to check who may announce an address
  - Certificates for servers, ways for a receiver to check where email came from

- Added-on security has a shape of its own, and you will see it every week:
  - Optional? Not everyone turns it on
  - Has to work alongside the old way, so the old way stays reachable

Wrap-up

Where the technical facts come from

- The protocols in this course are written down in documents called RFCs
  - RFC? a published specification for how an Internet protocol works
  - They are published by the IETF, the body that writes the Internet's standards

- When I tell you what a protocol does, that usually comes from its RFC

Summary

- You present twice
  - 30 minutes (weeks 5, 6, 8) and 60 minutes (weeks 10 to 15)
- Your questions count too
  - At least 2 in the first round, at least 4 in the second round
- Week 7 you submit a 20-minute video on your own research

- One story, four topics:
  - A name must be translated (DNS), the question must travel (BGP), the answer must prove itself (PKI and TLS), and email repeats all three
- None of these was designed with security
  - It was added later, it is optional, and the old way stays reachable